Project Design Phase-II

Streamlining Standard Laptop Orders with Flow Designer

Date	30.09.2026
Team ID	SWTID-2026-3846
Project Name	Streamlining standard Laptop orders with Flow Designer
Maximum Marks	4 Marks

1. Technical Architecture

The deliverable shall include the architectural diagram as below and the information as per the table 1 & table 2.

Example Order process during standard laptop orders for offline mode

Reference: https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/

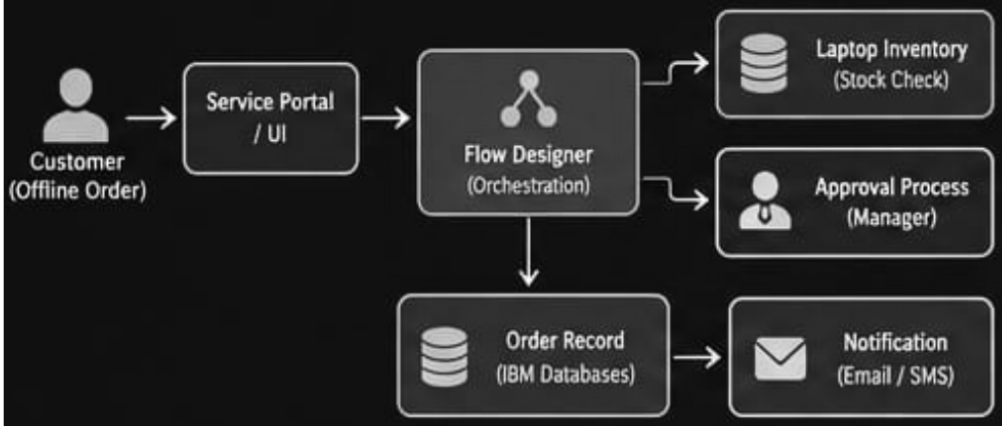


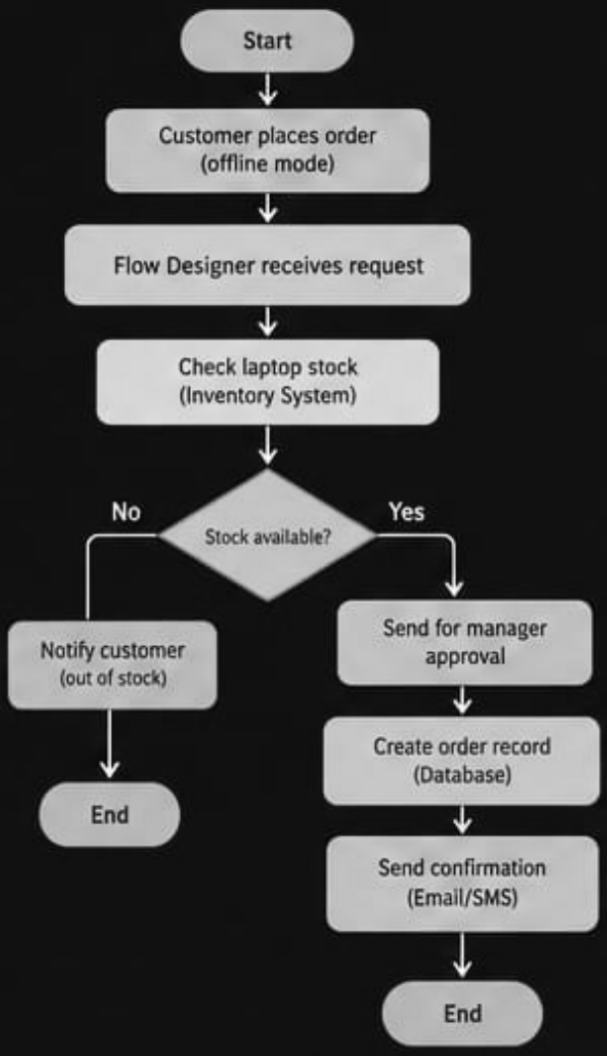
Table 1: Components & Technologies

S.No	Component	Description	Technology
1.	User Interface	How user interacts with application e.g., Web UI, Mobile App, Chatbot etc.	HTML, CSS, JavaScript Angular Js / React Js etc.
2.	Application Logic-1	Logic for a process in the application	Java / Python
3.	Application Logic-2	Logic for a process in the application	IBM Watson STT services
4.	Application Logic-3	Logic for a process in the application	IBM Watson Assistant
5.	Database	Data Type, Configurations etc.	MySQL, NoSQL etc.
6.	Cloud Database	Database Service on Cloud	IBM DB2, IBM Cloudant etc.
7.	File Storage	File storage requirements	IBM / block Storage or Other Storage Service of Local Filesystem
8.	External API-1	Purpose of External API used in the application	IBM Watson API, etc.
9.	External API-2	Purpose of External API used in the application	Aadhar API, etc.
10.	Machine Learning Model	Purpose of Machine Learning Model	Object Recognition Model, etc.
11.	Infrastructure (Server / Cloud)	Application Deployment on Local System / Cloud Local Server Configuration : Cloud Server Configuration :	Cloud Foundry, Kubernetes, etc.

Table 2: Application Characteristics

S.No	Characteristics	Description	Technology
1.	Open Source Frameworks	List the open-source frameworks used	Node.js, Python, Django
2.	Security Implementations	List all the security / access controls implemented, use of firewalls etc.	IAM, Firewall, Encryption
3.	Scalable Architecture	Justify the scalability of architecture (3-tier, Micro services)	Cloud Auto Scaling, Load Balancer
4.	Availability	Justify the availability of application (e.g. use of load balancers, distributed storage, etc.)	AWS / IBM Cloud, Multi-zone
5.	Performance	Design considerations for the performance of the application (number of requests per sec, use of Cache, use of CDN's etc.)	Redis Cache, CDN, Load Balancer

3. Flow Designer – Order Process Flow



4. User Stories

S.No	User Story
1.	As a customer, I want to place an order for a standard laptop offline, so that I can get the product without using the online portal.
2.	As a system, I want to check laptop stock availability, so that the order can be processed only if the item is available.
3.	As a manager, I want to receive an approval request, so that the order can be validated before confirmation.
4.	As a customer, I want to receive an email/SMS notification, so that I can track my order status.
3.	As an admin, I want to store order details in the database, so that it can be accessed for future reference and reporting.

5. Key Benefits

- Reduces manual effort and paperwork
- Faster order processing with Flow Designer
- Real-time stock verification
- Improved accuracy and transparency
- Better customer satisfaction

6. Challenges

S.No	Challenge	Description
1.	Real-time Stock Update	Inventory may not be updated instantly, leading to incorrect availability information.
2.	Approval Delays	Manual approval from managers may cause delays in order processing.
3.	Integration with External Systems	Connecting with ERP / external APIs can be complex and error-prone.
4.	Data Security	Handling sensitive customer and order data requires strong security measures.
5.	User Adoption	Employees may face difficulty in using the new flow if not trained properly.

7. Proposed Solution

S.No	Solution	Technology / Approach
1.	Real-time Inventory Integration	API integration with inventory system, Webhooks
2.	Automated Approval Workflow	Flow Designer with conditional logic and notifications
3.	API Connectors	IBM API Connect / REST APIs
4.	Security Controls	IAM, Encryption, Firewalls
5.	Training & Support	User guides, training sessions, help desk

8. Conclusion

The proposed solution using Flow Designer helps in automating the standard laptop order process, reducing manual work, improving accuracy and providing real-time updates. It ensures a smooth flow from order placement to delivery with proper approvals, stock checks and notifications, resulting in better customer satisfaction and operational efficiency.

9. References

1. https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/
2. https://www.ibm.com/cloud/architecture
3. https://aws.amazon.com/architecture/
4. https://www.ibm.com/docs/en/api-connect
5. https://www.ibm.com/cloud/flow-designer